

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA
Course: Web Technologies II (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Server? Compare monolithic and microservices architectures in terms of scalability, deployment, and maintenance. 7
 - b) Write a Python program to demonstrate the use of lists, tuples, and dictionaries with at least three operations on each. 8
 2. a) What is function in python? Write a Python program using functions to calculate the electricity bill as per the following conditions: 7
- For first 100 units – Rs. 5 per unit
For next 100 units – Rs. 7 per unit
Beyond 200 units – Rs. 10 per unit

OR

Write a Python code that accepts the details of 10 students (name, address, mobile number, and age) and writes these details into a CSV file.

- b) What is a framework, and why is it important for developing a secure application? Explain why flask is called as a micro framework? 8
3. a) What is a database? Why is SQLAlchemy used with Flask? Compare ORM and Core approaches in SQLAlchemy. 7
- b) Differentiate between sessions and cookies. Demonstrate sending email in flask with a suitable example. 8
4. a) What is a Content Management System (CMS)? Explain its features and advantages. Also, list out some popular CMS's. 7
- b) Write a Python Flask application that connects to a database using SQLAlchemy. The application should fetch all records from a *students* table and display them on an HTML page in tabular format. 8

OR

- What is SQLAlchemy? Explain its role in Flask applications. Write a sample Python code to demonstrate how to configure and connect a Flask application to a database using SQLAlchemy.
5. a) What is SOAP protocol? Describe the main HTTP methods used in RESTful APIs and their purposes. 7
 - b) Differentiate between development and production environments in web development. Why is it not recommended to run development servers in production? 8
 6. a) What do you mean by threat, vulnerabilities and attacks? Define injection and how can we be safe from it? 7
 - b) What is Web Application security? Explain the necessity of Role Based Access Control. Also, differentiate between Authentication and Authorization. 8
 7. Write short notes on: (**Any two**) 2×5
 - a) SSL/TLS
 - b) MVC Architecture
 - c) CSV and JSON